Gaussian Linear Systems with Structural Equality Constraints

Clarification of the Structural Model, Recovery of η_0, and the Case $A_j^p = A_j^q$

Consider the system

$$A_i \eta_0 = \sum_{j=1}^{n} A_j^i \eta_j^i, \qquad i = 1, \ldots, m,$$

and let

$$b_i := \sum_{j=1}^{n} A_j^i \eta_j^i, \qquad b := \mathrm{col}(b_1, \ldots, b_m), \qquad A_0 := \begin{bmatrix} A_1 \\ \vdots \\ A_m \end{bmatrix}.$$

Assume that A_0 has full column rank.

For the present discussion, all constraints are structural equalities. Thus when one writes

$$\eta_j^p = \eta_j^q,$$

this means that the two symbols denote the same latent draw from

$$\eta_j \sim \mathcal{N}(\mu_j, C_j).$$

On the other hand, if no equality identifies η_j^r with η_j^s, then those two symbols may represent independent observations with the same law $\mathcal{N}(\mu_j, C_j)$.

1. Structural model for shared and independent copies

A convenient way to encode the structural identities is to introduce, for each fixed j, an index map

$$\alpha(i, j)$$

that tells us which latent draw from family j is used in equation i. Then we introduce latent variables

$$\eta_{j,\alpha} \sim \mathcal{N}(\mu_j, C_j),$$

mutually independent across distinct pairs (j, α), and write

$$\eta_j^i = \eta_{j,\alpha(i,j)}.$$

This covers both cases:

- if $\alpha(p, j) = \alpha(q, j)$, then η_j^p and η_j^q are the same observation;
- if $\alpha(r, j) \neq \alpha(s, j)$, then η_j^r and η_j^s are independent observations, each distributed as $\mathcal{N}(\mu_j, C_j)$.

With this notation,

$$b_i = \sum_{j=1}^{n} A_j^i \eta_{j,\alpha(i,j)}.$$

Therefore

$$m_i := \mathbb{E}[b_i] = \sum_{j=1}^{n} A_j^i \mu_j, \qquad m := \mathrm{col}(m_1, \ldots, m_m).$$

The covariance between equations i and ℓ is

$$S_{i\ell} := \mathrm{Cov}(b_i, b_\ell) = \sum_{j=1}^{n} A_j^i C_j (A_j^\ell)^\top \mathbf{1}\{\alpha(i,j) = \alpha(\ell,j)\}.$$

Hence the stacked right-hand side satisfies

$$b \sim \mathcal{N}(m, S),$$

where S is built from the blocks $S_{i\ell}$ above.

Immediate consequence. The diagonal blocks are always

$$S_{ii} = \sum_{j=1}^{n} A_j^i C_j (A_j^i)^\top,$$

while off-diagonal blocks appear exactly when two equations share the same latent draw for at least one index j.

2. Recovering η_0: exact-transform form versus information form

The phrase "recovering η_0" can mean two different things.

2.1 Exact-transform form

This is the interpretation in which the stacked relation

$$A_0 \eta_0 = b$$

is assumed to hold exactly for every realization. In this case b must lie in $\mathrm{range}(A_0)$ almost surely.

Since A_0 has full column rank, it has a left inverse. A canonical choice is the Moore–Penrose left pseudoinverse

$$A_0^\dagger := (A_0^\top A_0)^{-1} A_0^\top,$$

which satisfies $A_0^\dagger A_0 = I$. More generally, any matrix L such that $LA_0 = I$ may be used. Then

$$\eta_0 = Lb.$$

Therefore, if

$$b \sim \mathcal{N}(m, S),$$

possibly with singular covariance S, one obtains

$$\eta_0 \sim \mathcal{N}(\mu_0, C_0), \qquad \mu_0 = Lm, \qquad C_0 = LSL^\top.$$

In particular, with $L = A_0^\dagger$,

$$\mu_0 = A_0^\dagger m, \qquad C_0 = A_0^\dagger S (A_0^\dagger)^\top.$$

This is the natural interpretation when the model is genuinely exact and η_0 is a real latent quantity determined by the shared observations.

2.2 Information form

A different interpretation treats b as Gaussian *information about* $A_0\eta_0$, rather than as an exactly compatible right-hand side. In that case one considers the generalized least-squares objective

$$\hat{\eta}_0(b) := \arg\min_{\eta}(b - A_0\eta)^\top S^{-1}(b - A_0\eta),$$

where S is assumed positive definite. The solution is

$$\hat{\eta}_0(b) = Kb, \qquad K := (A_0^\top S^{-1} A_0)^{-1} A_0^\top S^{-1}.$$

If b is random with law $\mathcal{N}(m, S)$, then the estimator $\hat{\eta}_0(b)$ is Gaussian with

$$\mathbb{E}[\hat{\eta}_0] = Km, \qquad \mathrm{Cov}(\hat{\eta}_0) = KSK^\top = (A_0^\top S^{-1} A_0)^{-1}.$$

So the formula

$$\mu_0 = (A_0^\top S^{-1} A_0)^{-1} A_0^\top S^{-1} m, \qquad C_0 = (A_0^\top S^{-1} A_0)^{-1}$$

should be read as the mean and covariance of the *estimator* $\hat{\eta}_0$, not automatically as the law of an exactly existing latent variable.

2.3 Relation between the two interpretations

The two viewpoints coincide only in special circumstances.

If the exact relation $b = A_0\eta_0$ really holds and $b \in \mathrm{range}(A_0)$ almost surely, then any left inverse L recovers η_0 exactly. Moreover, whenever the generalized least-squares map K is well-defined, one has

$$KA_0 = I,$$

so for every compatible right-hand side $b = A_0\eta_0$,

$$\hat{\eta}_0(b) = Kb = \eta_0.$$

Thus the two operators agree on the set of compatible data, even though they need not be the same matrix on the whole ambient space.

In many exact structural models, however, the covariance of b is singular because b lives on the lower-dimensional subspace $\mathrm{range}(A_0)$. In that situation the exact-transform form is usually the more natural description, whereas the information form is better interpreted as a noisy or approximate recovery rule.

3. The special case $A_j^p = A_j^q$

It is completely allowed to have identical coefficient matrices in different equations:

$$A_j^p = A_j^q.$$

What matters is whether the underlying observations are the same latent draw or independent draws.

For fixed j and two equations p, q,

$$\mathrm{Cov}(A_j^p \eta_j^p,\, A_j^q \eta_j^q) = A_j^p\, \mathrm{Cov}(\eta_j^p, \eta_j^q)\, (A_j^q)^\top.$$

If in addition

$$A_j^p = A_j^q =: A_j,$$

then two important subcases appear.

Same latent observation. If $\alpha(p,j) = \alpha(q,j)$, so that

$$\eta_j^p = \eta_j^q = \eta_{j,\alpha},$$

then

$$\mathrm{Cov}(A_j\eta_j^p, A_j\eta_j^q) = A_j C_j A_j^\top.$$

Independent observations with the same law. If $\alpha(p,j) \neq \alpha(q,j)$, then η_j^p and η_j^q are independent, and therefore

$$\mathrm{Cov}(A_j\eta_j^p, A_j\eta_j^q) = 0,$$

even though the coefficient matrices are identical.

Hence the statement

$$A_j^p = A_j^q$$

does *not* imply

$$\eta_j^p = \eta_j^q.$$

It only says that the same linear operator is applied in the two equations. Whether a cross-covariance appears is determined by the structural identity map $\alpha(i,j)$.

4. Summary for the current model

For the current structural-identity interpretation, the cleanest description is:

1. For each fixed j, equation i uses a latent draw $\eta_{j,\alpha(i,j)} \sim \mathcal{N}(\mu_j, C_j)$.
2. Distinct pairs (j,α) are independent.
3. The stacked right-hand side is

   $$b_i = \sum_{j=1}^n A_j^i \eta_{j,\alpha(i,j)}, \qquad b \sim \mathcal{N}(m,S),$$

   with

   $$m_i = \sum_{j=1}^n A_j^i \mu_j, \qquad S_{i\ell} = \sum_{j=1}^n A_j^i C_j (A_j^\ell)^\top \mathbf{1}\{\alpha(i,j) = \alpha(\ell,j)\}.$$

4. If the model is exact and $A_0\eta_0 = b$ holds almost surely, then the natural recovery is the exact-transform form

   $$\eta_0 = Lb, \qquad \mu_0 = Lm, \qquad C_0 = LSL^\top,$$

   for any left inverse L of A_0.
5. If instead b is treated as Gaussian information about $A_0\eta_0$ with positive definite covariance S, then the natural object is the generalized least-squares estimator

   $$\hat{\eta}_0(b) = (A_0^\top S^{-1} A_0)^{-1} A_0^\top S^{-1} b.$$

6. Equality of coefficient matrices, such as $A_j^p = A_j^q$, does not by itself force equality of the latent observations. Cross-covariances are controlled by the structural identities, not by coefficient equality alone.